Leave No User Behind: Towards Improving the Utility of Recommender Systems for Non-mainstream Users

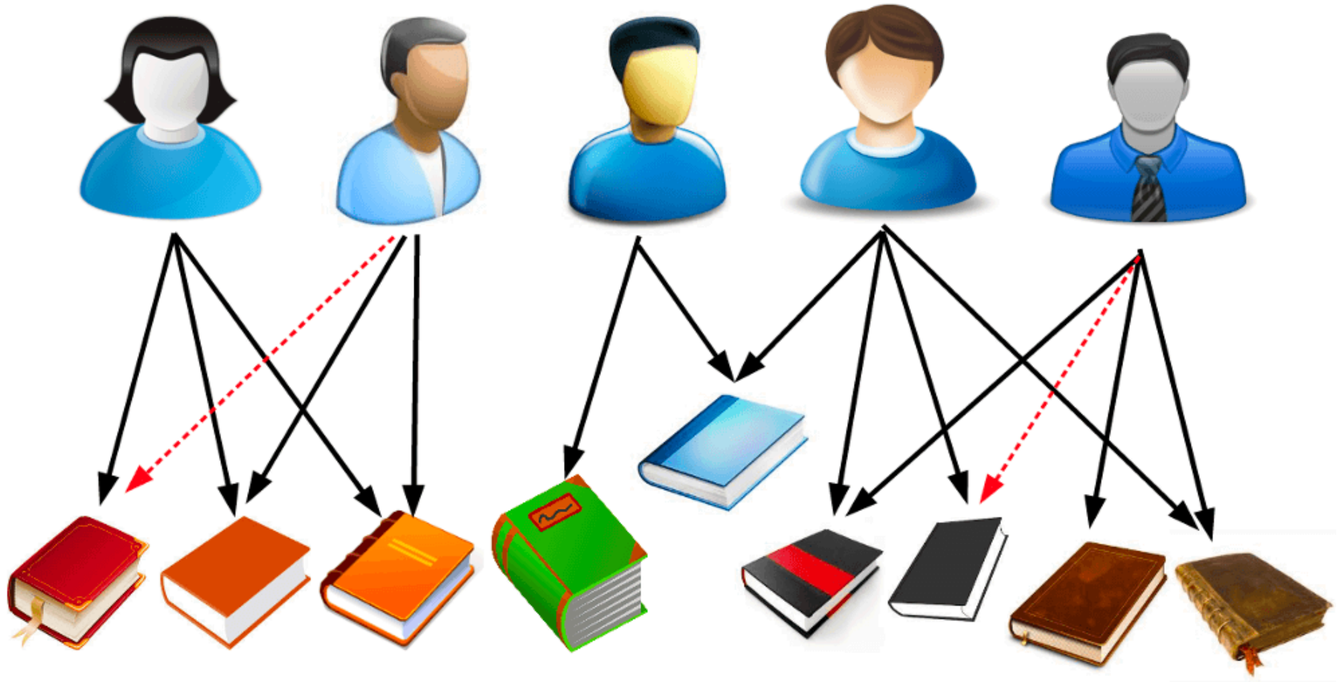
Roger Zhe Li (Presenter), Julián Urbano, Alan Hanjalic

Delft University of Technology, the Netherlands

Email: z.li-9@tudelft.nl

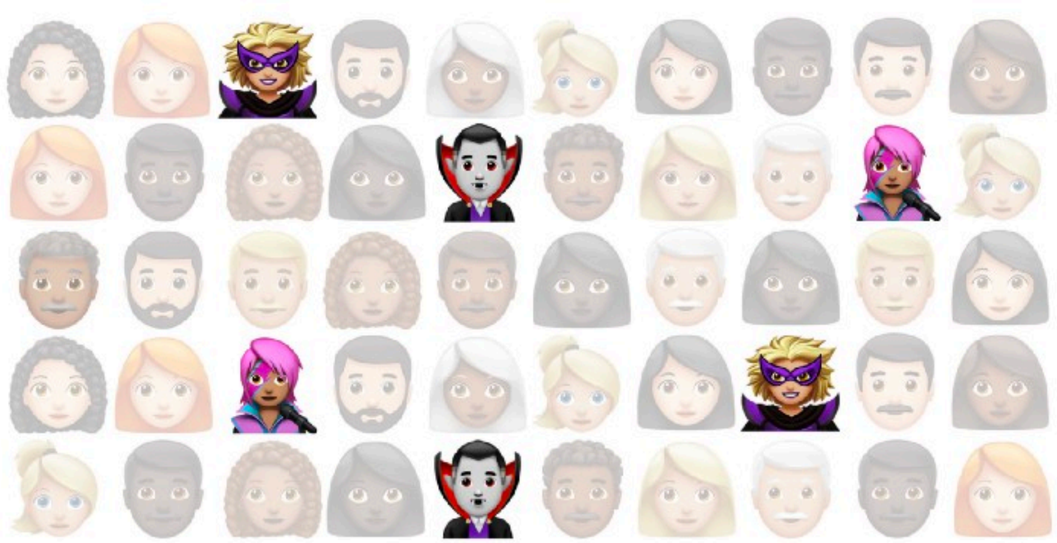
@Zhe_Delft

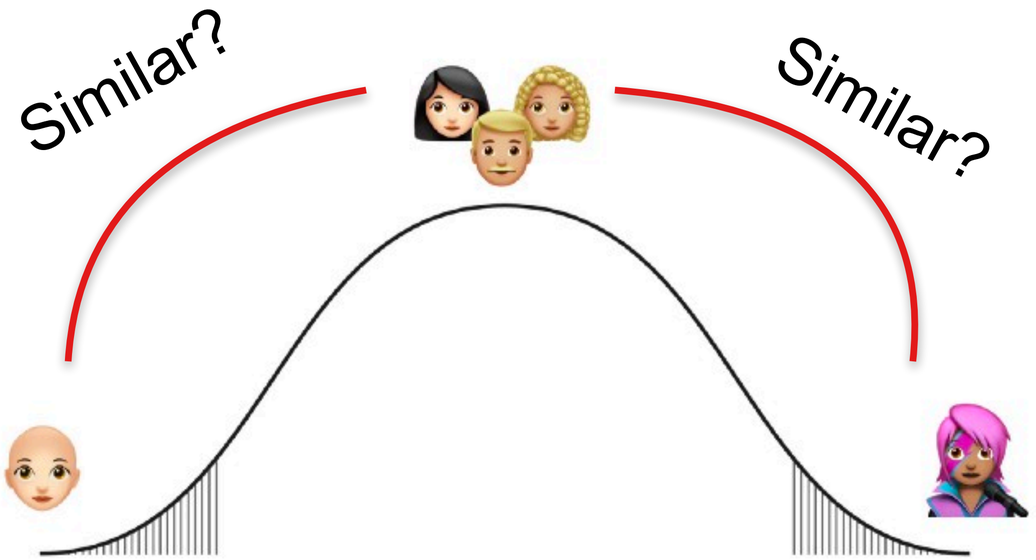
Motivation: Collaborative Filtering



CF is similarity-based.

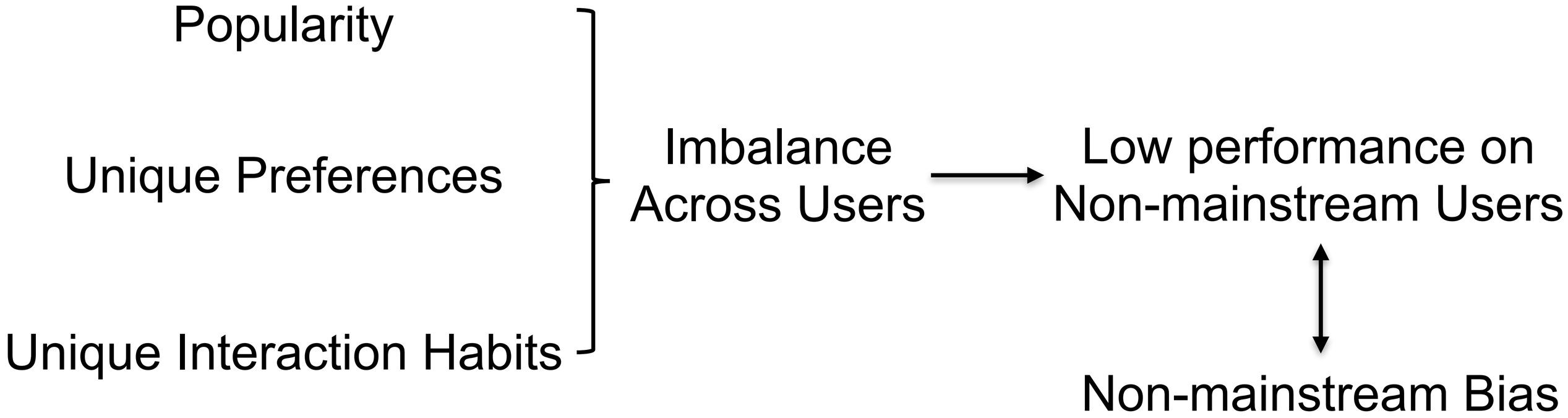
Similarity, Are We Going Too Far?





https://uxdesign.cc/the-fundamentals-of-engaging-with-extreme-users-45e0033e6b2

Biases in CF-based Recommendation



Strategies

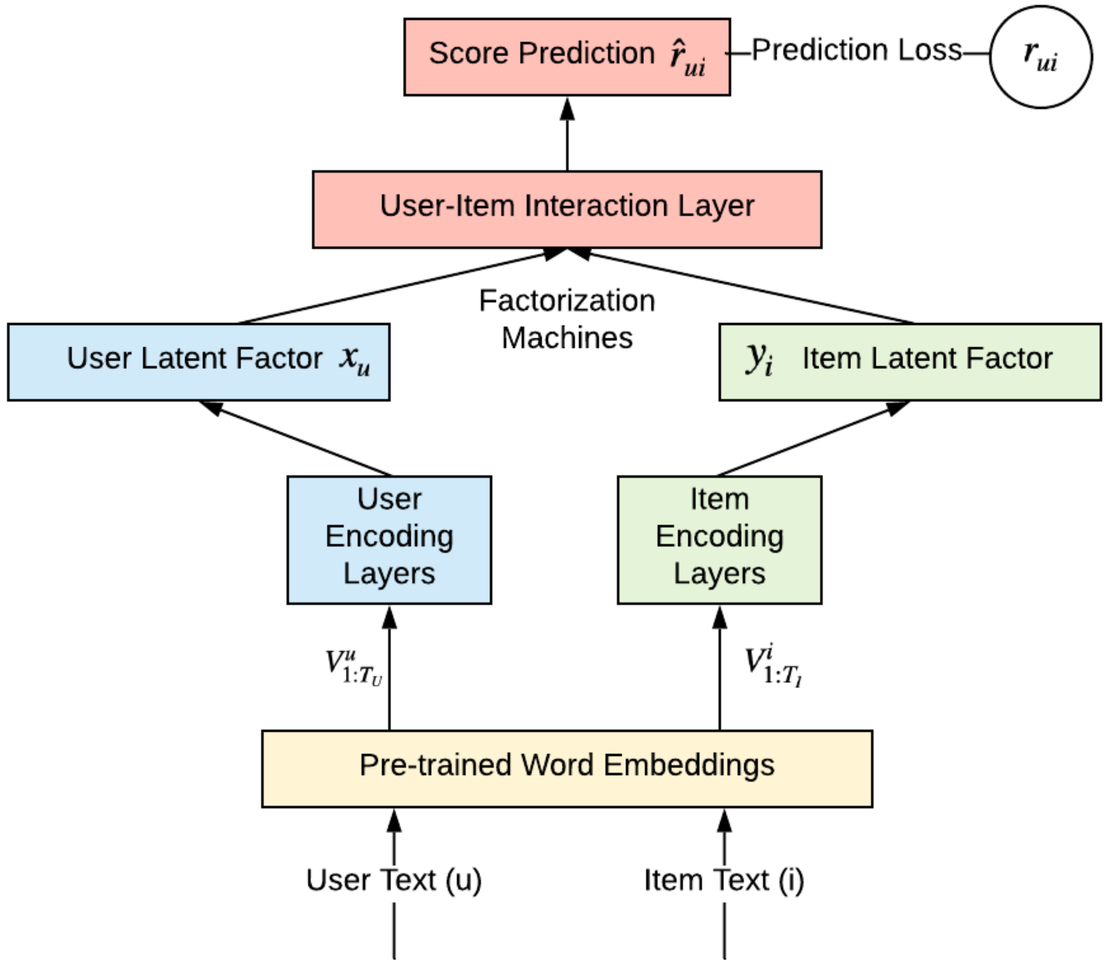
- Using content features to augment the information of non-mainstream users;
- Introducing an adversarial power to prevent pure similarity-oriented learning, and thus preserve the unique user and item properties.

Problem

- Goal: alleviating the non-mainstream bias.
- Given: explicit ratings; content-based side information (review texts).
- Target: A personalized recommendation list to each user, achieving a better balance across all users in recommendation.

How to utilize the content-based side information?

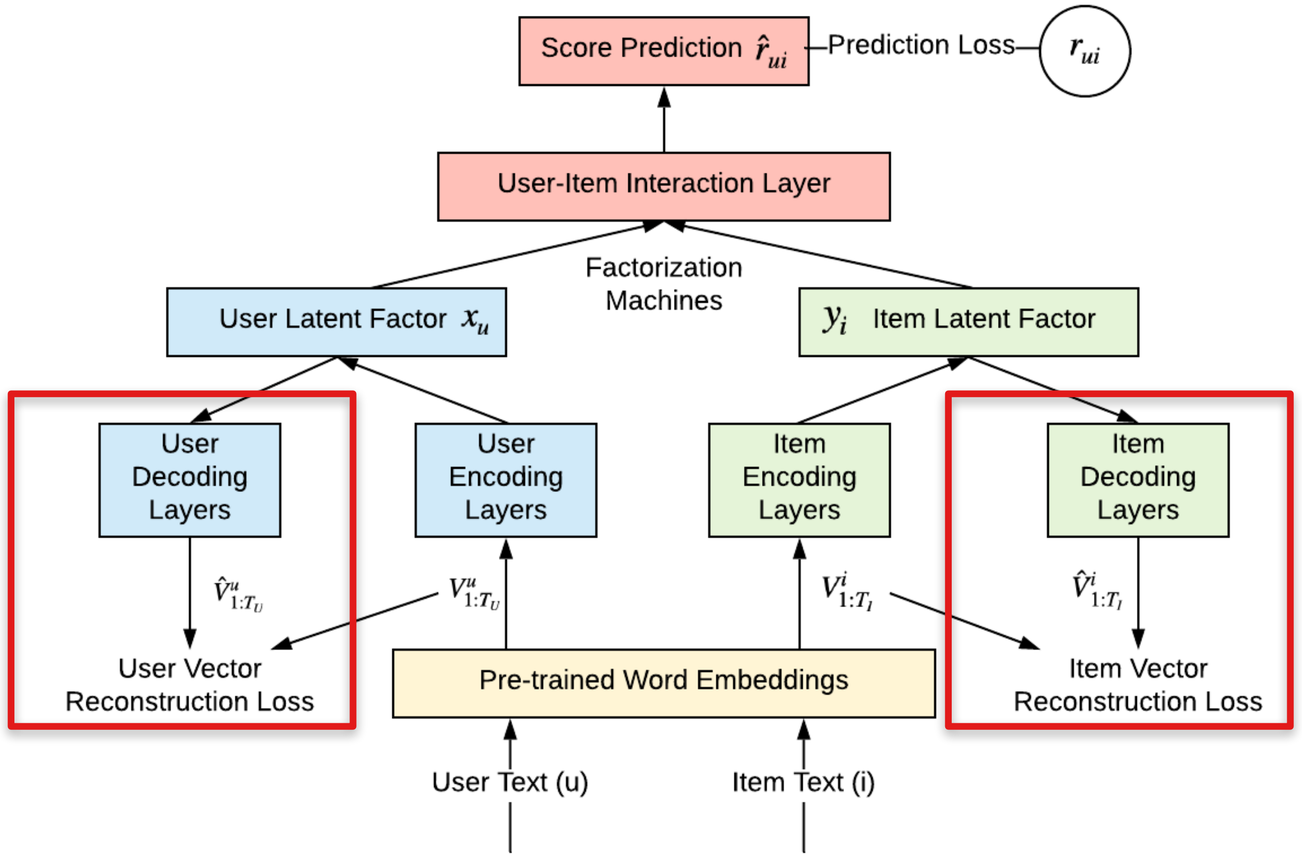
DeepCoNN: Preliminary Model



Zheng, Lei, Vahid Noroozi, and Philip S. Yu. "Joint deep modeling of users and items using reviews for recommendation." *Proceedings of the Tenth ACM International Conference on Web Search and Data Mining*. ACM, 2017

How to provide the adversarial power?

NAECF: Debiasing for Non-Mainstream Users



The architecture of *Neural AutoEncoder Collaborative Filtering*

Model Learning: Overall Loss

The overall loss function is the linear combination generated from CF (rating prediction) and AE (text reconstruction).

$$L = L_R + w\left(L_U + L_I\right)$$

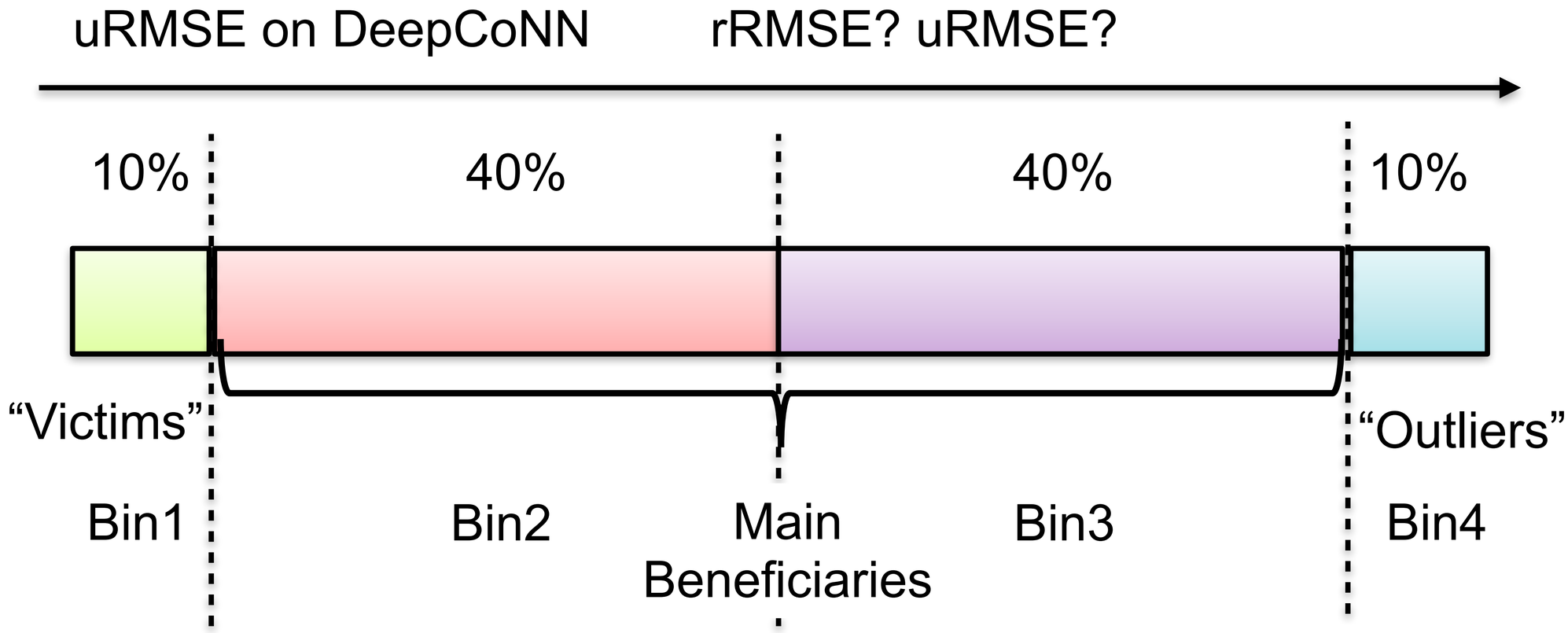
Trade-off between CF and AE

{0, 0.1, 0.2, 0.5, 1, 2, 5, 10}

Equivalent to DeepCoNN

How to evaluate the balance across users?

Individual User Investigation: AutoEncoders



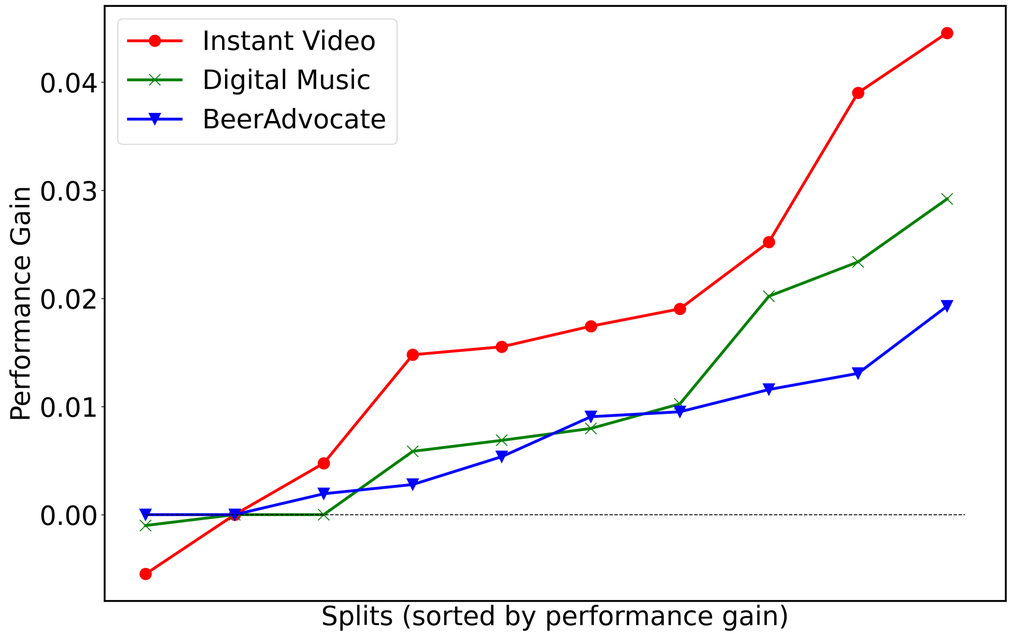
$$Performance\ Gain = 0.1 \cdot \Delta Bin1 + 0.4 \cdot \Delta Bin2 + 0.4 \cdot \Delta Bin3 + 0.1 \cdot \Delta Bin4$$

Experiments: Datasets & Baselines

Dataset	#users	#items	#ratings	Sparsity	#words
Instant videos	5,130	1,685	37,126	99.57	19M
Digital music	5,541	3,568	64,706	99.67	73M
BeerAdvocate	3,703	37,580	393,035	99.72	198M

- Train: Validation: Test = 8:1:1
- 10 different splits per dataset
- Core Evaluation Metric: rRMSE, uRMSE
- Each user has at least 5 ratings
- Baselines: MF and DeepCoNN

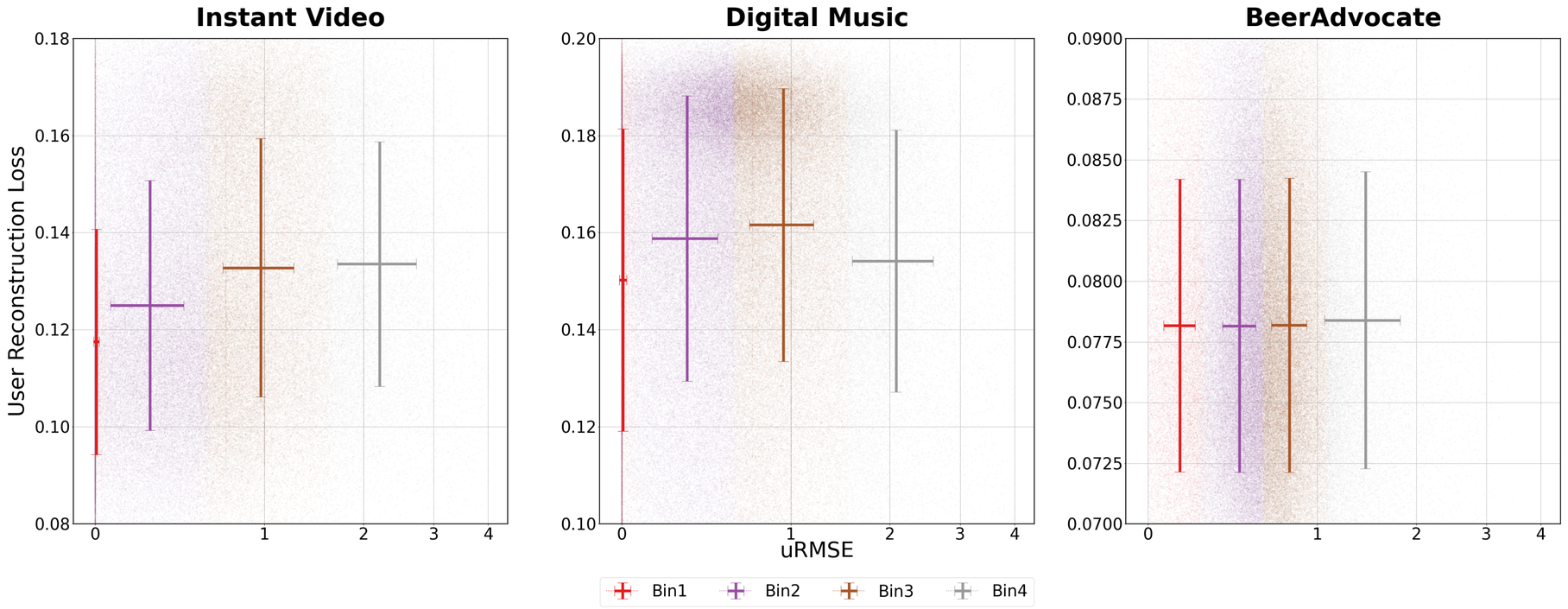
Individual User Investigation: AutoEncoders



- NAECF improves the utilities of non-mainstream users WITHOUT significantly hurting mainstream users.

Dataset	Δ_1	Δ_2	Δ_3	Δ_4	Δ
Instant Video	-0.0035	0.0256*	0.0267*	-0.0308*	0.0175*
Digital Music	0.0036	0.0184*	0.0106*	-0.0167*	0.0103*
BeerAdvocate	0.0119	0.0117*	0.0063*	-0.0115*	0.0073*

Individual User Investigation: Mechanisms



Dataset	Δ_1	Δ_2	Δ_3	Δ_4	Δ
Instant Video	-0.0035	0.0256*	0.0267*	-0.0308*	0.0175*
Digital Music	0.0036	0.0184*	0.0106*	-0.0167*	0.0103*
BeerAdvocate	0.0119	0.0117*	0.0063*	-0.0115*	0.0073*

- The vectors of Mainstream users are easier to reconstruct.

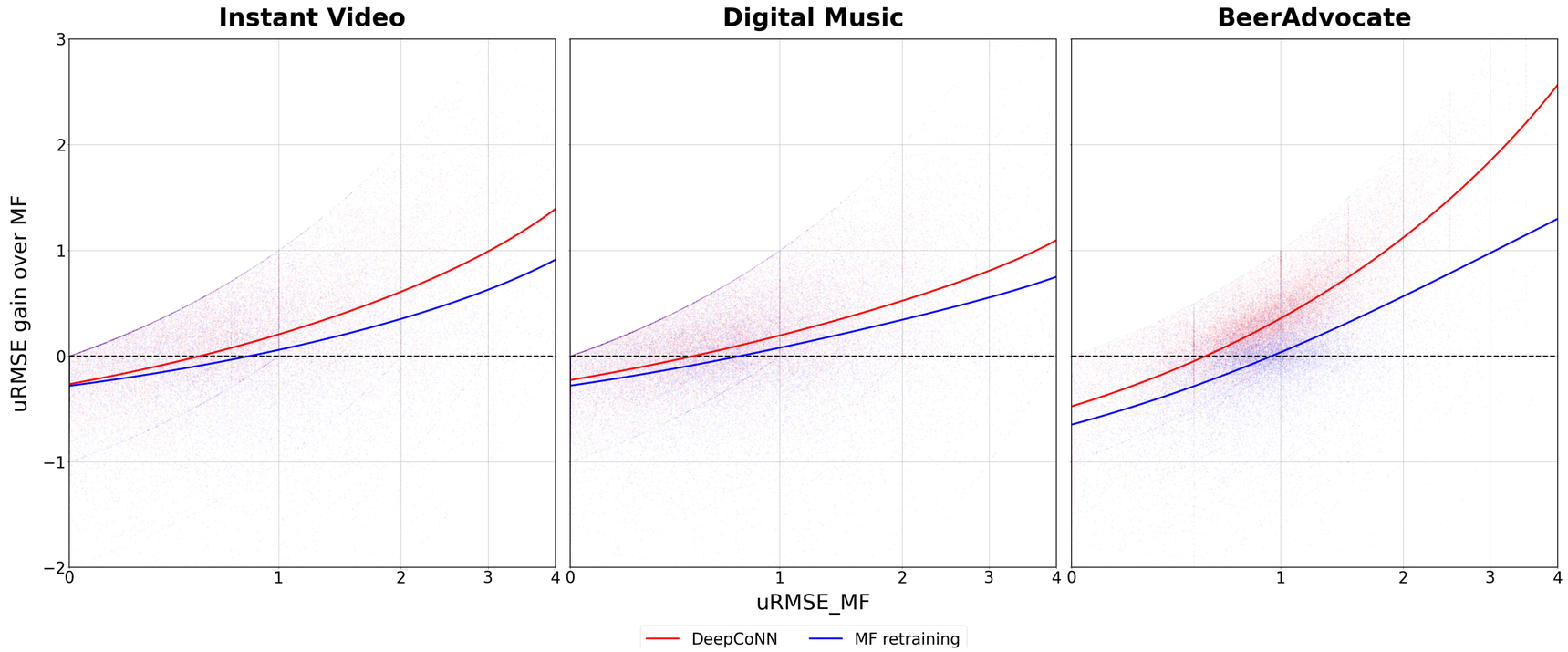
Experiments: Overall Results

Table 2: *rRMSE* over 10 data splits for all recommendation methods in all three datasets (*mean* ± *std.dev.*). Weighted NAECFs are represented by the weights of the text rebuilding loss *w*. Best results per dataset are in bold. Results statistically different from the best (*t*-test, $p < 0.05$) are marked with *.

Dataset	MF	DeepCoNN	NAECF						
			0.1	0.2	0.5	1.0	2.0	5.0	10.0
Instant Video	1.1600 ± .0264*	0.9744 ± .0145	**0.9732 ± .0149**	0.9749 ± .0122	0.9754 ± .0159	0.9757 ± .0169	0.9798 ± .0212	0.9896 ± .0221*	0.9967 ± .0221*
Digital Music	1.0466 ± .0097*	**0.9078 ± .0138**	0.9083 ± .0115	0.9106 ± .0128	0.9097 ± .0114	0.9104 ± .0128	0.9118 ± .0134	0.9167 ± .0108	0.9219 ± .0146*
BeerAdvocate	1.0442 ± .0048*	0.6722 ± .0090	0.6707 ± .0064	**0.6692 ± .0035**	0.6746 ± .0059*	0.6756 ± .0082*	0.6785 ± .0098*	0.6899 ± .0137*	0.7068 ± .0278*

- Review features help achieve a significant improvement over pure MF.
- NAECF with a mild CF-AE trade-off does NOT bring a significant rRMSE loss.

Individual User Investigation: Reviews



- Review features help achieve a significant improvement over pure MF on user balance.

Conclusions

- Collaborative filtering methods suffer a non-mainstream bias across different users;
- With the adversarial power of AutoEncoders, NAECF achieves a better user balance on top of DeepCoNN.
- Reviews can help promote the recommendation utility for non-mainstream users;

Code & Data: https://github.com/roger-zhe-li/wsdm21-mainstream

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